

Digital Electronics (NECE 102)

(State diagram and Timing diagram of Flip flop)

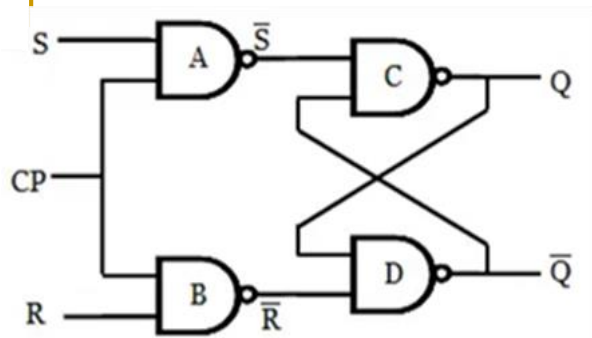
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The state diagram

- It is the pictorial representation of the behavior of sequential circuits.
 - It is a pictorial representation of the relationships between the present state, the input state, the next state, and the output state of a sequential circuit.
 - It clearly shows the transition of states from the present state to the next state and output for a corresponding input.
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SR Flip Flop



Truth Table

CLK	S	R	Q	$\bar{Q}$
0	X	X	Hold	
1	0	0	Hold	
1	0	1	0	1
1	1	0	1	0
1	1	1	Invalid	

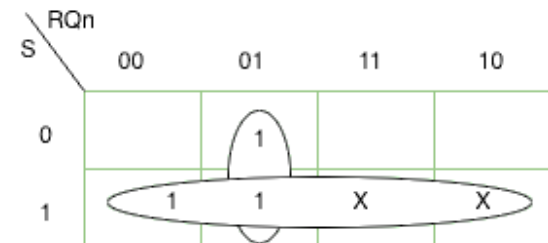
CLK	S	R	Qn+1	State
0	X	X	Qn	Hold
1	0	0	Qn	Hold
1	0	1	0	Reset
1	1	0	1	Set
1	1	1	X	Invalid

Characteristic table

S	R	Qn	Qn+1
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	Invalid
1	1	1	Invalid

Excitation table

Qn	Qn+1	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0



$$Q_{n+1} = S + R' Q_n$$

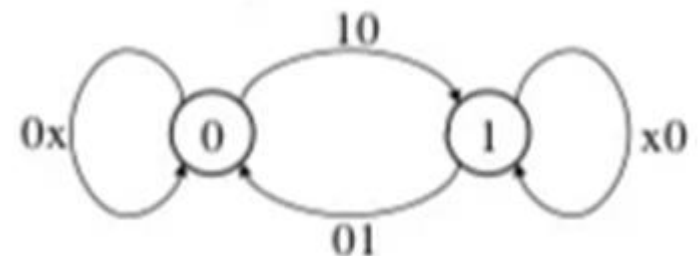
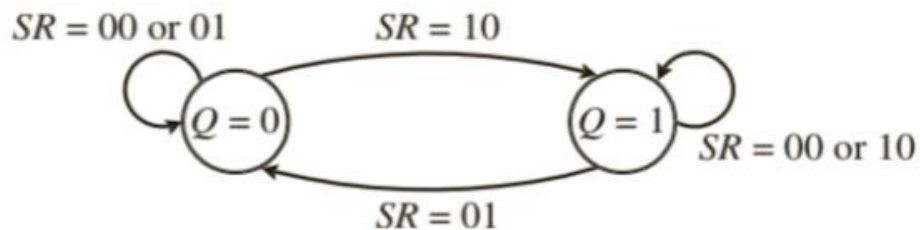
SR Flip Flop: State diagram

Characteristic table

S	R	Q _n	Q _{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	Invalid
1	1	1	Invalid

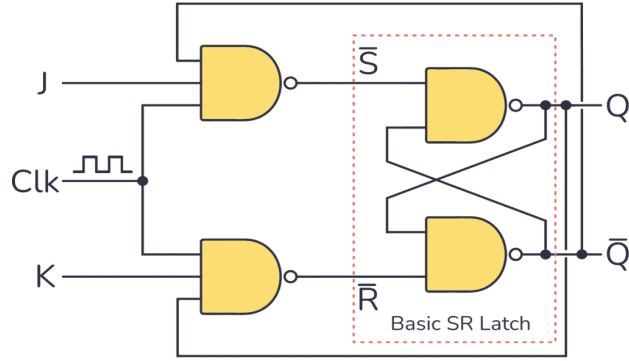
Excitation Table

Q _n	Q _{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0



$$Q_{next} = S + R'Q$$

J K Flip Flop - State diagram



Truth Table

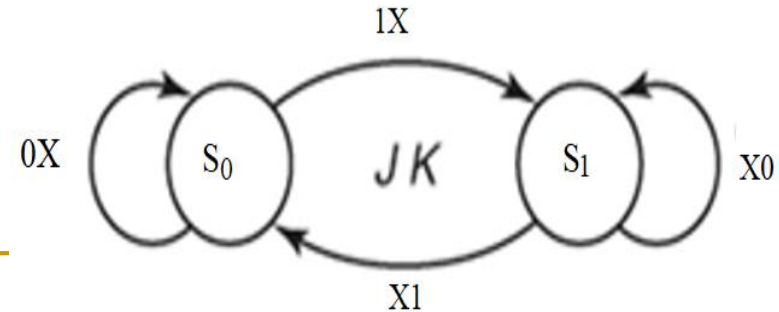
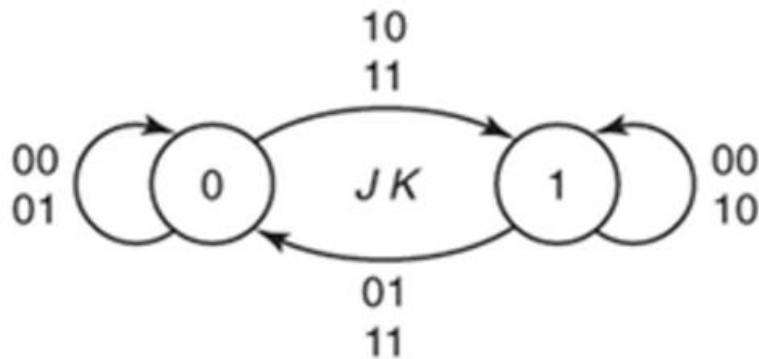
Clock	J	K	Q_{n+1}	State
0	x	x	Q_n	Hold
1	0	0	Q_n	Hold
1	0	1	0	Reset
1	1	0	1	Set
1	1	1	$\bar{Q}_n$	Toggle

Characteristic table

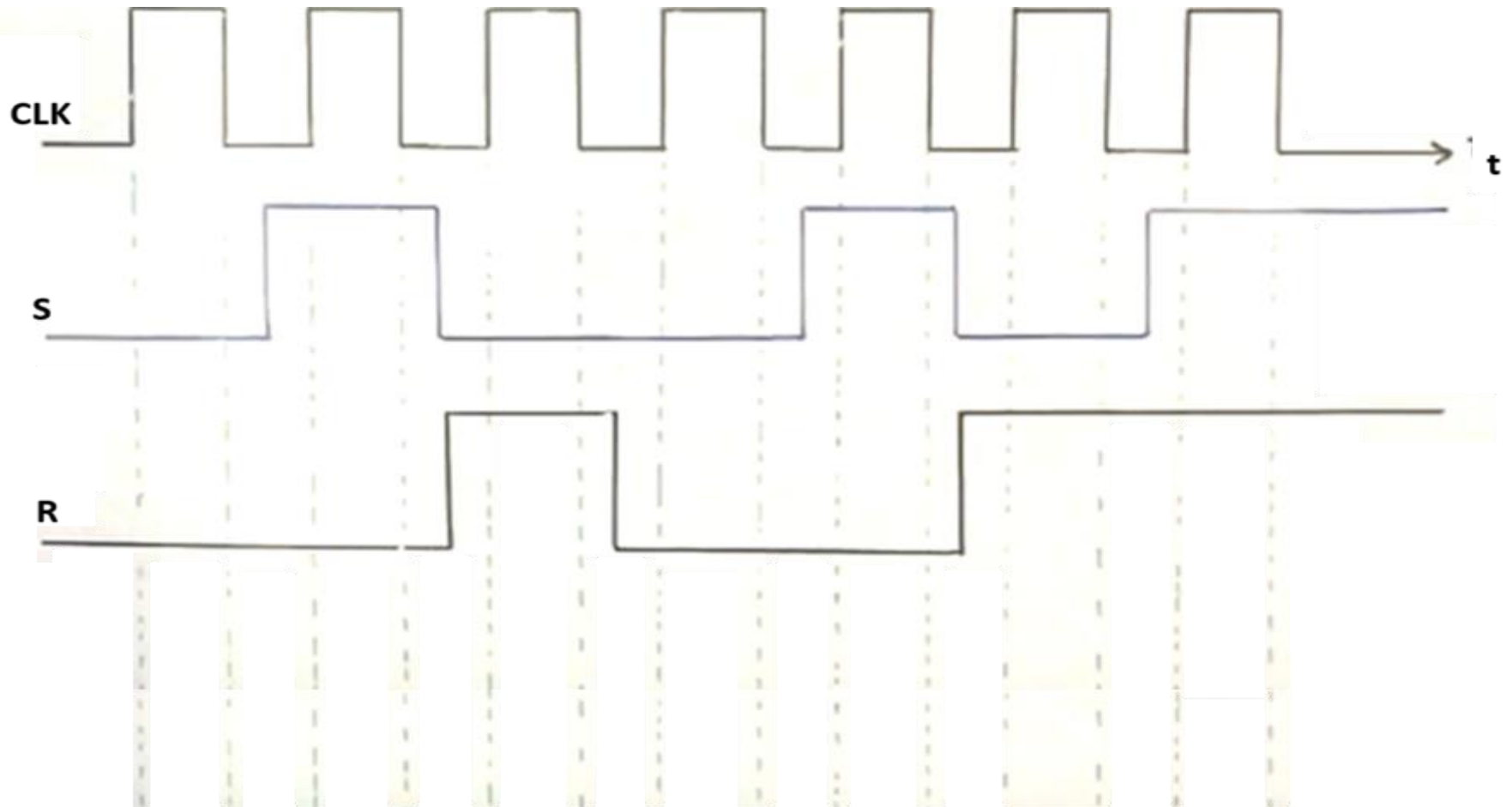
INPUT			OUTPUT
J	K	Q_n	$Q(n+1)$
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	0

Excitation table

INPUT		OUTPUT	
Q_n	$Q(n+1)$	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

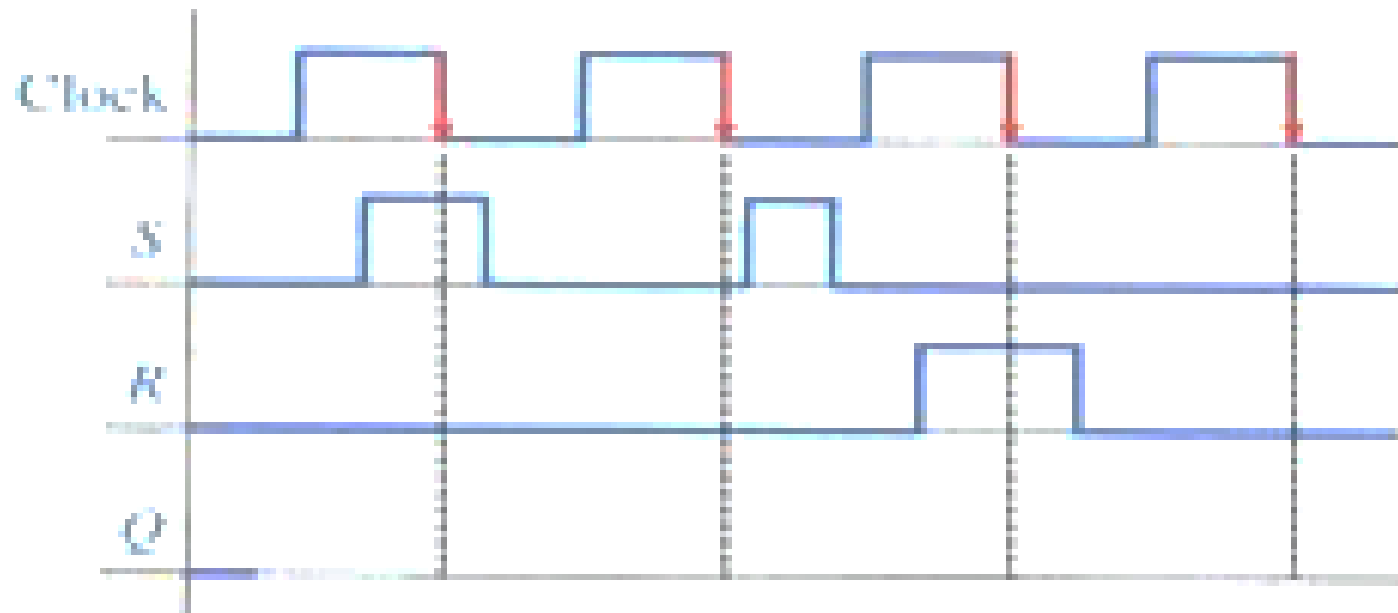


Timing diagram of SR Flip Flop



Timing diagram of SR Flip Flop

- Fill in the timing diagram for a falling-edge triggered S-R flip-flop.
 - Assume Q begins at 0



Timing diagram for JK Flip-flop

Negative Edge Triggered

